COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures & Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	7/17/26	**DATE SUBMITTED:**	7/17/26
NAME:	Gabriel Dwyane D. Arididdon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 2.1 **Hands-on Activity 2.1 Arrays, Pointers and Dynamic Memory Allocation**

1. PROCEDURE OUTPUT

What procedures did you perform?

Screenshot

```cpp
#include <iostream>
#include <string>

class Student {
private:
    std::string studentName;
    int studentAge;

public:
    // constructor
    Student(std::string newName = "Gabo Duweyn", int newAge = 19) {
        studentName = newName;
        studentAge = newAge;
        std::cout << "The constructor is called\n";
    }

    // destructor
    ~Student() {
        std::cout << "The destructor is called\n";
    }

    // copy constructor
    Student(const Student& copyStudent) {
        std::cout << "The copy constructor is called\n";
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

    // copy assignment operator
    Student& operator=(const Student& copyStudent) {
        std::cout << "The copy assignment operator is called\n";
        if (this != &copyStudent) {
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }
        return *this;
```

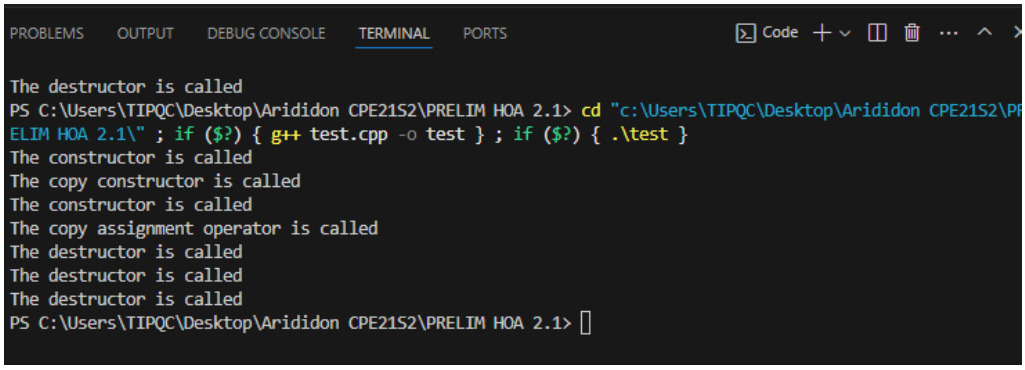
<table>
<tr>
<td></td>
<td>


```
    // display attributes
    void printDetails() {
        std::cout << studentName << " " << studentAge << std::endl;
    }
};

int main() {
    Student student1("Thomas, 29");
    Student student2(student1);
    Student student3;
    student3 = student2;

    return 0;

}
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS        Code

The destructor is called
PS C:\Users\TIPQC\Desktop\Arididdon CPE21S2\PRELIM HOA 2.1> cd "c:\Users\TIPQC\Desktop\Arididdon CPE21S2\PR
ELIM HOA 2.1\" ; if ($?) { g++ test.cpp -o test } ; if ($?) { .\test }
The constructor is called
The copy constructor is called
The constructor is called
The copy assignment operator is called
The destructor is called
The destructor is called
The destructor is called
PS C:\Users\TIPQC\Desktop\Arididdon CPE21S2\PRELIM HOA 2.1>
```
</td>
</tr>
<tr>
<td>Observation</td>
<td>When doing the copy assignment operator, errors occurred because delete requires the object to be on dynamic memory allocation. Since std::string and int are stored automatically (stack-based), they don’t need manual delete.</td>
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Table 2-1 Initial Drive Program

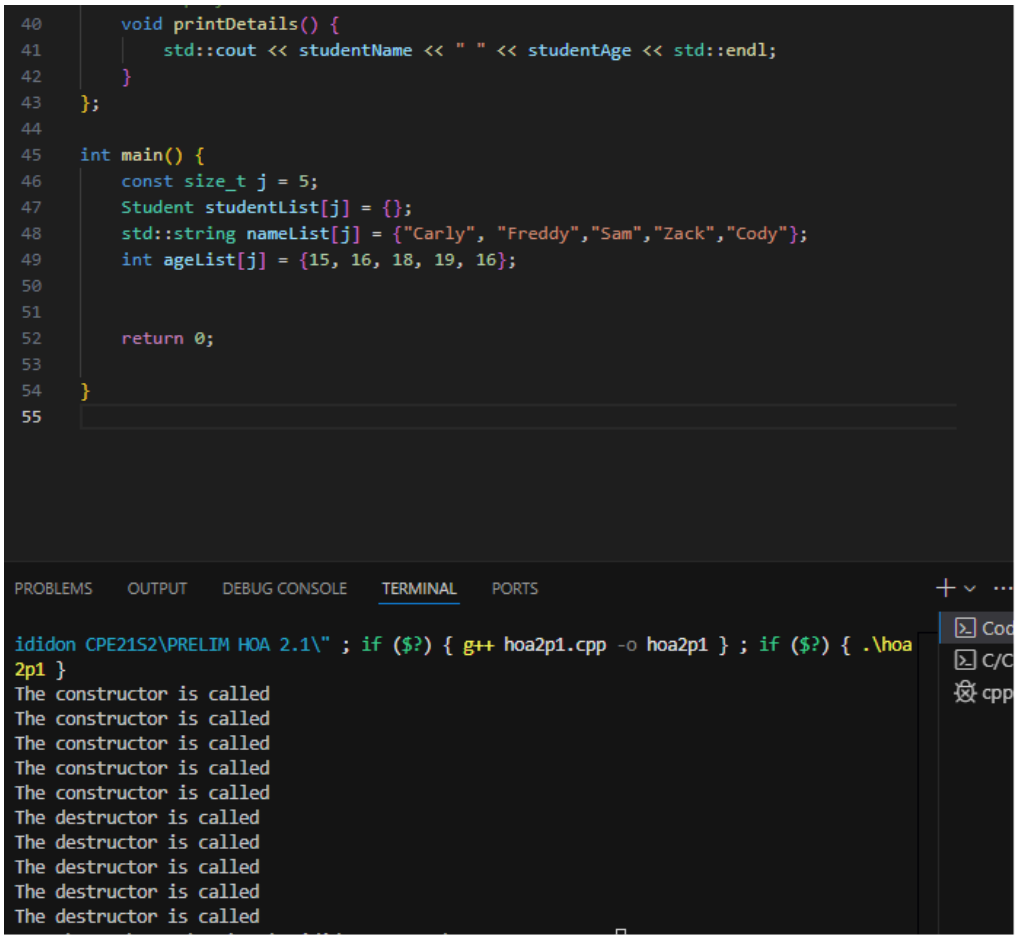
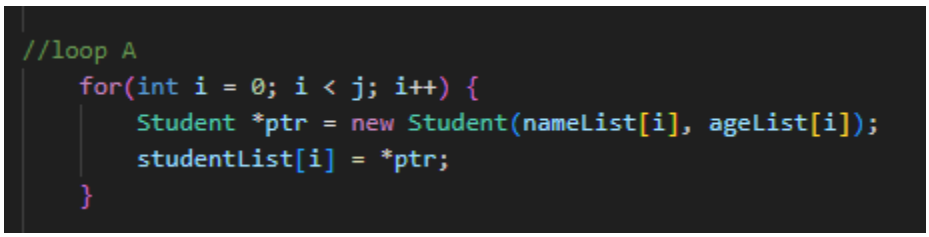
Screenshot	
Observation	After running the code, the constructor was called five times, once for each element. At the end of the program, the destructor was also called five times, showing that it automatically handles object creation and destruction for arrays.

Table 2-2 Modified Driver Program with Student Lists

Loop A	
Observation	Based on what I've observed loop A iterates through names and ages and in each index, it dynamically creates a Student object then it is then assigned to a studentList. The constructor is called when the object is created, the copy assignment is called when it is assigned in the list

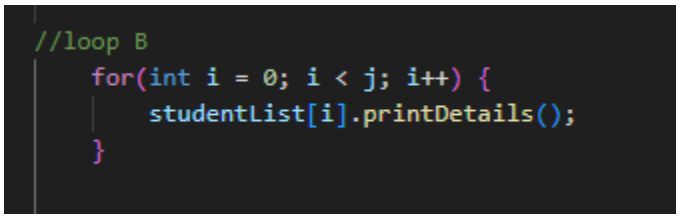
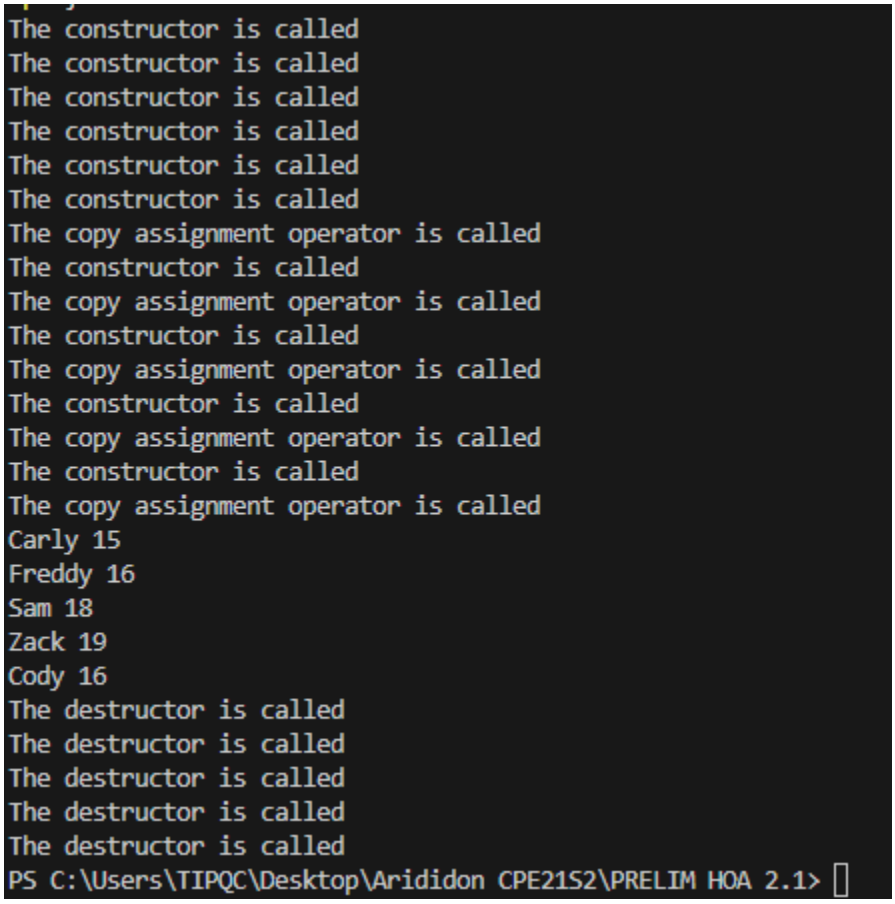
	then, the temporary object is destroyed, so the destructor is also called.
Loop B	
Observation	On loop B, it simply loops and prints the details of each studentList in each index. The constructor, copy assignment, and destructor are also called in each index. This displays the stored name and age for each student, confirming that the data was successfully copied into the array.
Output	
Observation	The output shows multiple calls to the constructor, copy assignment operator, and destructor during Loop A and Loop B. It also prints the correct student details (Carly 15, Freddy 16, Sam 18, Zack 19, Cody 16), showing that the assignment worked and the attributes were properly stored.

Table 2-3 Final Driver Program

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve programming problems using dynamic memory allocation, arrays and pointers

Jenna's Grocery List		
Apple	PHP 10	x7
Banana	PHP 10	x8
Broccoli	PHP 60	x12
Lettuce	PHP 50	x10

Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.

Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.

Screenshot

```cpp
#include <iostream>
#include <string.h>

class Fruit {
private:
    std::string* fruitName;
    float* fruitPrice;
    int* fruitQuantity;

public:
    //constructor
    Fruit(std::string newName = "Apple", float newPrice = 10, int newQuantity = 7 ) {
        fruitName = new std::string(newName);
        fruitPrice = new float(newPrice);
        fruitQuantity = new int(newQuantity);
        std::cout << "Constructor called\n";

    }
    //destructor
    ~Fruit() {
        delete fruitName;
        delete fruitPrice;
        delete fruitQuantity;
        std::cout << "Destructor called\n";
    }

    //copy constructor
    Fruit(const Fruit& copyFruit) {
        std::cout << "Copy Constructor called\n";
        fruitName = copyFruit.fruitName;
        fruitPrice = copyFruit.fruitPrice;
        fruitQuantity = copyFruit.fruitQuantity;
```

```
    }
    //copy assignment operator
    Fruit& operator=(const Fruit& copyFruit) {
        std::cout << "Copy Assignment Operator called\n";

            if (this != &copyFruit) {

                delete fruitName;
                delete fruitPrice;
                delete fruitQuantity;

                fruitName = new std::string(*copyFruit.fruitName);
                fruitPrice = new float(*copyFruit.fruitPrice);
                fruitQuantity = new int(*copyFruit.fruitQuantity);
            }

        return *this;
    }
```

```
//calculate sum
float calculateSum() {
    return (*fruitPrice) * (*fruitQuantity);
}
```

```
    //display attributes
    void display() {
        std::cout << *fruitName << " " << "PHP"
                  << " " << *fruitPrice << " " << "x"
                  << *fruitQuantity << "\n";
    }
};
```

```cpp
class Veggie {
private:
    std::string* veggieName;
    float* veggiePrice;
    int* veggieQuantity;

public:

    //constructor
    Veggie(std::string newName = "Broccoli", float newPrice = 60, int newQuantity = 12 ) {
        veggieName = new std::string(newName);
        veggiePrice = new float(newPrice);
        veggieQuantity = new int(newQuantity);
        std::cout << "Constructor called\n";

    }
    //destructor
    ~Veggie() {
        delete veggieName;
        delete veggiePrice;
        delete veggieQuantity;
        std::cout << "Destructor called\n";
    }

    //copy constructor
    Veggie(const Veggie& copyVeggie) {
        std::cout << "Copy Constructor called\n";
        veggieName = copyVeggie.veggieName;
        veggiePrice = copyVeggie.veggiePrice;
        veggieQuantity = copyVeggie.veggieQuantity;
    }
```

```cpp
//copy assignment operator
Veggie& operator=(const Veggie& copyVeggie) {
    std::cout << "Copy Assignment Operator called\n";

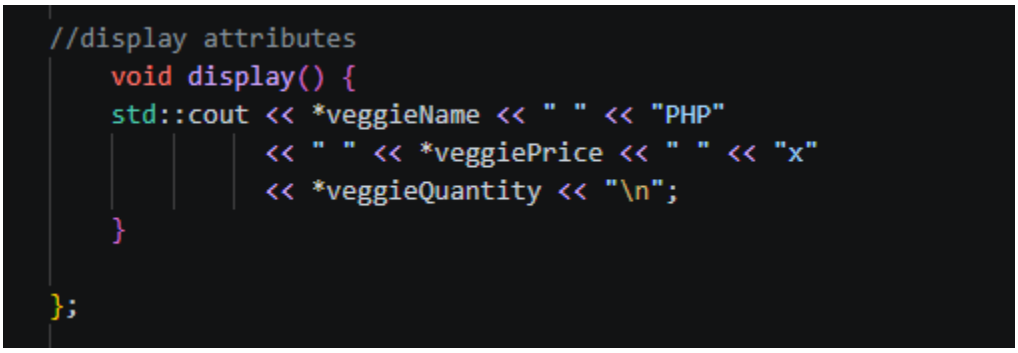
        if (this != &copyVeggie) {

            delete veggieName;
            delete veggiePrice;
            delete veggieQuantity;

            veggieName = new std::string(*copyVeggie.veggieName);
            veggiePrice = new float(*copyVeggie.veggiePrice);
            veggieQuantity = new int(*copyVeggie.veggieQuantity);
        }

    return *this;
}
```

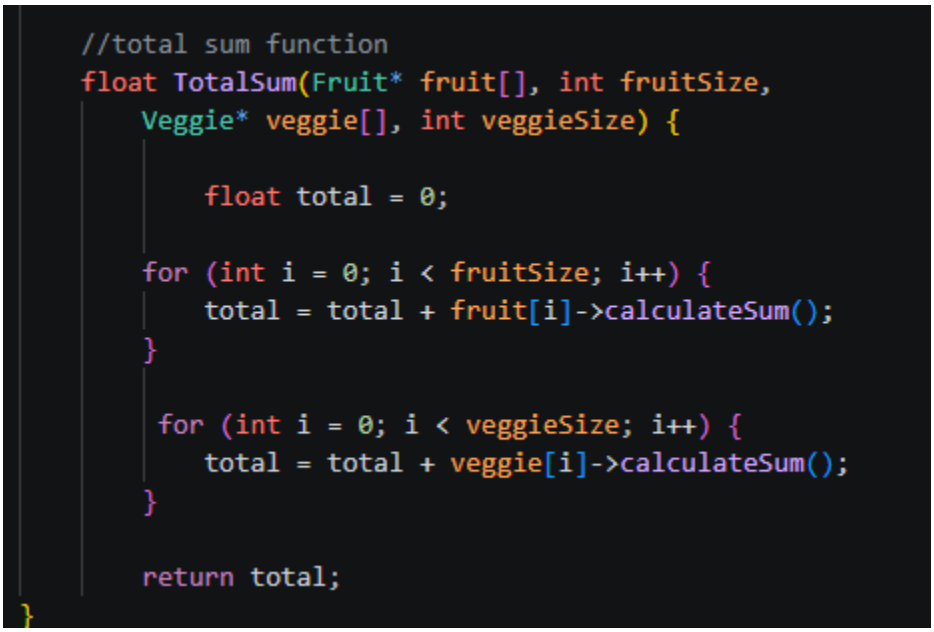
```cpp
//calculate sum
    float calculateSum() {
    return (*veggiePrice) * (*veggieQuantity);
    }
```

	
Observation	I created separate classes for fruits and vegetables to distinguish between the two types of grocery items. For the constructor, destructor, copy constructor, and copy assignment operator, I applied the concepts from the previous task but modified the attributes to use pointers (*) and dynamic memory allocation through new. Since the data is dynamically allocated, I used delete in the destructor to properly release the allocated memory and prevent memory leaks. I also adjusted the copy constructor and copy assignment operator to perform deep copying so that each object has its own separate memory allocation. It also has calculateSum function that multiplies the price to its quantity.

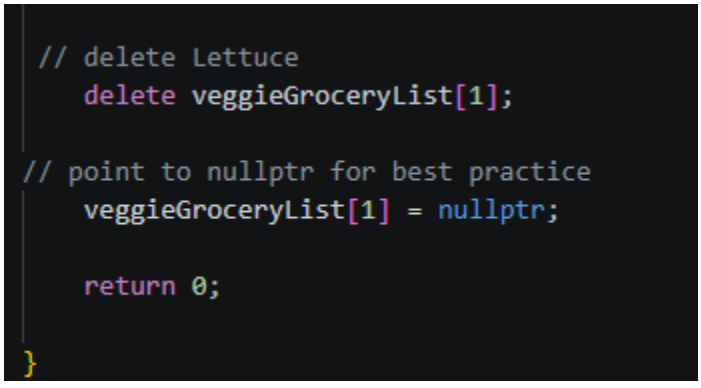
Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.

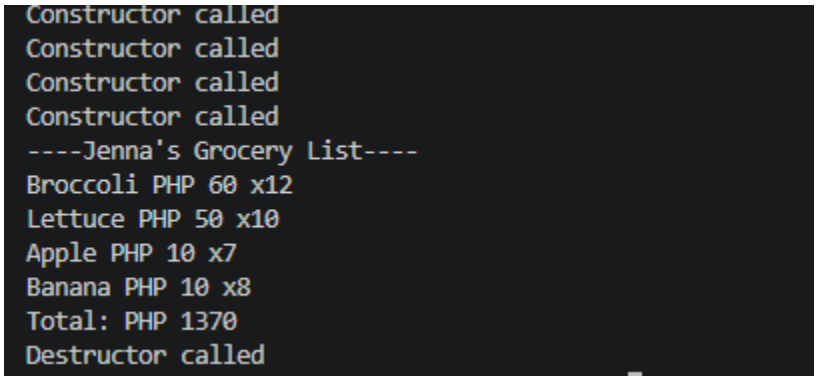
Screenshot	
Observation	I created two separate grocery list arrays for fruits and vegetables since they are stored as different object types. I used separate loops to iterate through each array and access every item during each iteration. Since the arrays contain object pointers, I used the -> operator to access the object's functions and display the details of each grocery item.

Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in Jenna's Grocery List.

Screenshot	
Observation	I created a TotalSum function that calculates the total cost of all items in Jenna's Grocery List. The function accepts the fruit and vegetable arrays along with their respective sizes as parameters. Using separate loops, it accesses each object in the arrays and calls the calculateSum() function using the -> operator to compute the subtotal of each item based on its price and quantity. The subtotals are then added together using the total variable, and the final value is returned as the overall grocery cost.

Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned

Screenshot	
Observation	I removed the Lettuce object from the grocery list using delete to deallocate the dynamically allocated memory. I then assigned the pointer to nullptr as a safety measure to prevent it from pointing to freed memory.

Output	
Observation	The output shows that four objects were successfully created, as indicated by the four constructor calls. The grocery list was displayed by accessing each stored object and showing its details. The TotalSum() function successfully calculated the total cost of all items, resulting in PHP 1370. The destructor was called after deleting the Lettuce object using the delete operator, which properly deallocated the memory assigned to that object.

Provide the following:

- Summary of lessons learned
- Analysis of the procedure
- Analysis of the supplementary activity
- Concluding statement / Feedback: How well did you think you did in this activity? What are your areas
- for improvement?

In this activity, I learned how to create and use classes in C++ while applying concepts such as constructors, destructors, copy constructors, copy assignment operators, pointers, and dynamic memory allocation. I was able to understand how objects are stored, accessed, and managed using arrays and pointers, as well as how functions like TotalSum() can be used to process multiple objects. However, I still need more practice with loops, proper pointer allocation, and using nullptr as a good practice in memory management. I also found the TotalSum() function and dynamic allocation challenging at first, but through this activity, I gained a better understanding of how they work. Overall, I think I did well, but I need to continue practicing these concepts to become more comfortable and improve.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Copilot, ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Assistance in error fixing, reviewing concepts of loops, guide for copy assignment operator, and dynamic allocation.**

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Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used Copilot and ChatGPT to have guidance when I can no longer fix the error by myself. I also learned that it is not needed to use delete when the object is not dynamically allocated and the code will continue since it is automatically deleted in stack allocation. I used them as a guide to better understand the concepts and assistance in debugging errors, reviewing concepts related to loops, guide in creating copy assignment operator, and clarifying the process of dynamic memory allocation.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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